SOFTWARE ENGINEERING LAB 4

Vibe Coding – Chat History

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Project Name	Traffic Escape – Pygame
Lab	Software Engineering Lab – Lab 4
Repository	PES1UG24AM350/50_traffic-escape

This document records the prompts used during the Vibe Coding process for the four implemented tasks, along with the implementation direction and testing outcomes.

TASK 1 – IMPLEMENT LIVES SYSTEM

USER – QUESTION / PROMPT

Implement a lives system in Traffic Escape. The player should start with 3 lives. When the player collides with a vehicle, decrease the number of lives by one and reset the player to the starting position while lives remain. Display the remaining lives on the HUD. When all lives are used, end the game and show the Game Over state. Preserve the existing collision handling and make sure a collision and a win cannot be registered in the same update.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

The game engine was updated to track three player lives, display the remaining lives on the HUD, and respawn the player after a collision while lives remain. When the lives reach zero, the game enters the Game Over state. The collision flow also stops immediately after a collision so the win condition cannot be set during the same update.

USER – TESTING / RESULT

The game was tested by colliding with traffic repeatedly. A collision reduced the life count, the player returned to the starting position while lives remained, and the game ended after the third life was lost. The collision/win state conflict was also addressed. Task 1 was confirmed working.

TASK 2 – ADD MOVING LOG / RAFT LANE

USER – QUESTION / PROMPT

Add a moving log or raft lane to Traffic Escape so the player can cross the water safely by jumping onto a moving log. The player must be able to land on a log, remain on it, and move with it horizontally. If the player is in the water without standing on a log, the player should lose a life after a short grace period. Keep the existing traffic, lives system, and other game behaviour working.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

A moving-log lane was added to the game. Logs travel horizontally across the water, and the player is carried along when standing on a log. A short grace period is used to avoid instant failure while the player is landing or transitioning. If the player remains in the water without support, a life is lost and the player is reset according to the lives system.

USER – TESTING / RESULT

The game was tested by jumping onto moving logs, riding them across the water, and checking the unsupported-water condition. The player could board and travel with a log, while falling into the water without support triggered the expected life-loss behaviour. Task 2 was confirmed working.

TASK 3 – ADD HIGH SCORE TABLE

USER – QUESTION / PROMPT

Add a persistent high-score table to Traffic Escape. Track the top 5 scores and save them to a JSON file so they remain available after the game is closed and reopened. Update the table when a run ends, avoid saving the same run more than once, and display the top scores on the appropriate game-over or end screen. Preserve the lives and moving-log features.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

A high-score module was added to load and save score data in JSON format. The game engine records a completed run once and keeps the highest five scores. The top-five table is shown on the end screen, and the JSON file allows the scores to persist across sessions.

USER – TESTING / RESULT

The game was tested by completing runs and checking that scores were recorded and displayed in descending order. The saved JSON data was retained between sessions, and the same run was not added repeatedly. Task 3 was confirmed working.

TASK 4 – ADD DAY / NIGHT CYCLE

USER – QUESTION / PROMPT

Add a day/night cycle to Traffic Escape. Switch between a day appearance and a darker night appearance at a short interval suitable for a lab demonstration. During night, make vehicles visible with headlights or light beams and show DAY or NIGHT on the HUD. Keep all previous features—including lives, moving logs, and the high-score table—working. Use a 10-second transition interval so the change can be observed during a short game.

LLM – IMPLEMENTATION RESPONSE / DIRECTION

A day/night cycle was added with a 10-second interval. The game changes its environment between day and night, displays the current state on the HUD, and uses vehicle headlights or light beams to improve visibility at night. The existing gameplay features were retained.

USER – TESTING / RESULT

The game was run long enough to observe the day-to-night and night-to-day transitions. The DAY/NIGHT HUD indicator changed with the scene, and vehicle lights were visible during the night state. The earlier lives, moving-log, and high-score functionality was retained. Task 4 was confirmed working.

FINAL PROJECT STATUS

Task	Feature	Status
Task 1	Lives System + Collision/Win State Fix	Completed
Task 2	Moving Log / Raft Lane	Completed
Task 3	Persistent Top 5 High Scores	Completed
Task 4	Day/Night Cycle + Vehicle Lights	Completed

Final deliverables: before-change video, after-change video, updated source code pushed to the personal GitHub repository, separate commits for the tasks, and this Vibe Coding chat history document.

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